# Kai Yao

Trustworthy and Secure AI
PhD Candidate, School of Informatics, University of Edinburgh, UK
**Expected PhD Completion: Oct 2026**
kai.yao@ed.ac.uk | kaiyao17@gmail.com | +86 17302151562 | +44 7867 763608
kaikaiyao.github.io | linkedin.com/in/kai-yao-ai-algorithm

## Profile

PhD candidate in Cyber Security, Privacy and Trust at the University of Edinburgh, specializing in secure and trustworthy machine learning, generative-image provenance, model attribution, adversarial robustness, privacy, and fairness. My research combines empirical evaluation, threat modelling, statistical measurement, and ML systems engineering. Previously, I worked as an AI algorithm engineer at Intel and Huawei, developing AI software components, optimization workflows, and performance benchmarks for production ML systems. My earlier work includes peer-reviewed research in computer vision for scientific and biomedical imaging at Johns Hopkins University.

## Research Areas

AI security and trustworthy ML; generated-image provenance and model attribution; AI fingerprinting and watermarking; adversarial robustness; privacy and fairness measurement; computer vision for scientific imaging; ML systems, quantization, and performance evaluation.

## Education

- **University of Edinburgh** — 2023 – Oct 2026 expected
  *PhD in Cyber Security, Privacy and Trust* — Edinburgh, UK
  - Advisor: Dr. Marc Juarez. Research focus: secure and trustworthy ML, generative-model provenance, AI image fingerprinting, adversarial robustness, privacy, and fairness.
  - Funded by a full PhD scholarship from the University of Edinburgh.
- **Johns Hopkins University** — 2020
  *MS in Mechanical Engineering* — Baltimore, MD, USA
  - Research focus: computer vision, quantitative microscopy, single-cell imaging, and deep learning for scientific measurement.
- **Fudan University** — 2017
  *BS in Theoretical and Applied Mechanics* — Shanghai, China
- **Osaka University** — 2014 – 2015
  *One-year undergraduate exchange student* — Osaka, Japan

## Research Experience

- **Generative AI Provenance, Model Attribution, and Post-Deployment Auditing** — 2024 – 2026
  *University of Edinburgh* — Edinburgh, UK
  - Developed methods and evaluation protocols for attributing generated images to source models, auditing model changes after certification, and assessing provenance reliability under adversarial manipulation.
  - **SPRINT:** designed a keyed passive-attribution method that uses verifier-held secret pixel-reconstruction targets. The method trains source-specific reconstructors from black-box generator outputs, keeps the verification task private from adaptive attackers, and reduces adaptive removal and forgery success while maintaining high clean attribution accuracy.
  - **FARE:** developed an output-only auditing method for detecting bait-and-switch image generators after certification. The method learns a forensic acceptance region from the certified generator and flags model-version, checkpoint, and inference-pipeline changes from generated outputs.
  - Built reproducible pipelines for generator sampling, source-specific training, attack simulation, fixed-FPR thresholding, AUROC/TPR/FPR analysis, prevalence-aware precision-recall analysis, and ablation studies.
  - Analyzed deployment implications for digital forensics, provider accountability, regulatory compliance, intellectual-property protection, and content authenticity.
- **Smudged Fingerprints: Robustness of AI Image Fingerprinting** — 2024 – 2025
  *University of Edinburgh* — Edinburgh, UK
  - Conducted a systematic security evaluation of AI image fingerprinting methods for generated-image attribution under adaptive manipulation.
  - Formalized white-box and black-box threat models for two attack goals: fingerprint removal, which erases identifying traces to evade attribution, and fingerprint forgery, which induces misattribution to a target model.
  - Implemented five removal and forgery attack strategies and benchmarked 14 representative fingerprinting methods across RGB, frequency, and learned-feature domains on 12 state-of-the-art GAN, VAE, and diffusion generators.
  - Measured clean and attacked performance, documented utility-robustness trade-offs, and identified evaluation assumptions that can overstate forensic reliability under adaptive attacks.
- **Differential Privacy's Disparate Impact in Machine Learning** — 2023 – 2024
  *University of Edinburgh* — Edinburgh, UK
  - Studied how differentially private training changes group-level error allocation across privacy mechanisms, model architectures, and data distributions.
  - Developed a taxonomy of disparity-inducing factors and identified dataset size, group imbalance, and group distance to decision boundaries as drivers of DP-induced unfairness.
  - Analyzed privacy-induced fairness degradation as a measurement problem, separating average utility loss from subgroup-specific harms and identifying when private learning changes the allocation of errors.

- **Deep Learning for Single-Cell Imaging and Quantitative Biology** 2018 – 2020
  *Johns Hopkins University* Baltimore, MD, USA
  - Developed label-free deep learning algorithms for microscopy-image cell classification, single-cell volume prediction, and morphology-driven phenotyping.
  - Built experimental and computational pipelines spanning microfluidic fluorescence-exclusion measurement, DIC microscopy, image preprocessing, U-Net/CNN model training, and validation against biological measurements.
  - Co-first author on studies in *Journal of Cell Science*, *Scientific Reports*, and *Bio-protocol* that developed computer-vision tools for quantitative biology.
- **Computational Modelling for Biomedical Mechanics** 2015 – 2017
  *Fudan University and collaborators* Shanghai, China
  - Contributed to phantom-based experimental validation of fast virtual deployment of self-expandable stents for cerebral aneurysm applications.
  - Applied mechanics, modelling, and validation methods to biomedical-engineering problems at the interface of computation and experimental measurement.

## Industry Experience

- **Senior AI Algorithm Engineer** 2021 – 2023
  *Intel Corp.* Shanghai, China
  - Developed and optimized Intel AI software components, including Neural Coder, Intel Extension for PyTorch, and Intel Neural Compressor, to improve PyTorch and TensorFlow training/inference performance on Intel hardware.
  - Developed PyTorch adapter and model-optimization workflows for Intel Neural Compressor, supporting INT8 quantization and accelerated inference for large generative-AI workloads, including Stable Diffusion.
  - Designed and maintained benchmarking frameworks to compare AI workload performance across Intel and non-Intel platforms, diagnose bottlenecks, and guide software optimization priorities.
  - Collaborated with partner teams including Alibaba Cloud and AWS to integrate Intel AI optimizations into production environments and translate benchmark results into engineering recommendations.
  - Received Intel division awards for Fast Stable Diffusion on Intel CPU, Neural Coder innovation, and Neural Coder collaboration with Alibaba Cloud.
- **AI Algorithm Engineer** 2020 – 2021
  *Huawei Technologies Co., Ltd.* Shanghai, China
  - Developed ML algorithms for 5G MU-MIMO systems using CNN and RNN models under practical deployment constraints.
  - Optimized model architectures and compressed model weights to improve training and inference efficiency, working with deployment and validation teams.

## Publications and Manuscripts

**Peer-Reviewed Publications in AI and Machine Learning**

- **Yao K**, Juarez M. *Smudged Fingerprints: A Systematic Evaluation of the Robustness of AI Image Fingerprints.* To appear in the 4th IEEE Conference on Secure and Trustworthy Machine Learning (SaTML), 2026.
- **Yao K**. *What Does the Credential Still Certify? Cognitive Stewardship for AI-Mediated Education.* To appear in the 9th AAAI/ACM Conference on AI, Ethics, and Society (AIES), 2026.
- **Yao K**. *When AI Does the Work, What Is Learning For? Post-Instrumental Learning and the Risk of Capacity Dissolution.* To appear in the 9th AAAI/ACM Conference on AI, Ethics, and Society (AIES), 2026.
- **Yao K**, Juarez M. *SoK: What Makes Private Learning Unfair?* Proceedings of the 3rd IEEE Conference on Secure and Trustworthy Machine Learning (SaTML), 2025.

**Peer-Reviewed Publications in AI for Science and Biomedical Engineering**

- Rochman ND*, **Yao K***, Gonzalez NA*, Wirtz D, Sun SX. *Single-Cell Volume Measurement Utilizing the Fluorescence Exclusion Method (FXm). Bio-protocol.* 2020;10(12):e3652.
- **Yao K***, Rochman ND*, Sun SX. *CTRL: A Label-Free Artificial Intelligence Method for Dynamic Measurement of Single-Cell Volume. Journal of Cell Science.* 2020;133(7):jcs245050.
- Perez-Gonzalez NA*, Rochman ND*, **Yao K***, Tao J, Le MT, Flanary S, Sablich L, Toler B, Crentsil E, Takaesu F, Lambrus B. *YAP and TAZ Regulate Cell Volume. Journal of Cell Biology.* 2019;218(10):3472–3488.
- **Yao K***, Rochman ND*, Sun SX. *Cell Type Classification and Unsupervised Morphological Phenotyping from Low-Resolution Images Using Deep Learning. Scientific Reports.* 2019;9(1):1–13.
- Zhang Q, Meng Z, Zhang Y, **Yao K**, Liu J, Zhang Y, Jing L, Yang X, Paliwal N, Meng H, Wang S. *Phantom-Based Experimental Validation of Fast Virtual Deployment of Self-Expandable Stents for Cerebral Aneurysms. BioMedical Engineering OnLine.* 2016;15(2):431–437.

*Note: * denotes equal contribution / co-first authorship.*

**Preprints and Manuscripts Under Review**

- **Yao K**, Juarez M. *SPRINT: Robust Model Attribution of Generated Images via Secret Pixel Reconstruction.* arXiv:2508.05691, 2026.

## Selected Talks, Presentations, and Media Coverage

- **Yao K**. *Smudged Fingerprints: A Systematic Evaluation of the Robustness of AI Image Fingerprints.* Oral presentation, IEEE Conference on Secure and Trustworthy Machine Learning (SaTML) 2026
- **Yao K**. *Fingerprinting AI Models in a Black-Box Setting.* Invited keynote, Kraus & Lederer Law Firm, on AI model fingerprinting and intellectual-property protection 2026
- **Yao K**. *Image Provenance in the AI Era.* Guest lecture, graduate course Privacy and Security with Machine Learning, University of Edinburgh 2026
- **Yao K**. *SoK: What Makes Private Learning Unfair?* Oral presentation, IEEE Conference on Secure and Trustworthy Machine Learning (SaTML) 2025
- **Yao K**. *Algorithmic Unfairness in Private Classifiers.* University of Edinburgh Cyber Security, Privacy, & Trust Research Showcase 2024
- **Media coverage:** University of Edinburgh News, DIGIT, and Herald Scotland 2026

## Teaching Experience

- **Guest Lecturer**, graduate-level Privacy and Security with Machine Learning, University of Edinburgh 2025 – 2026
  - Delivered lecture material on image provenance, AI fingerprinting, privacy, security, and trustworthy ML.
- **Teaching Assistant and Lab Demonstrator**, Privacy and Security with Machine Learning, University of Edinburgh 2023 – 2025
  - Supported labs, tutorials, assessment, and student questions on privacy, security, attacks, defences, and evaluation.
- **Teaching Assistant**, Mathematical and Computational Image Analysis, Johns Hopkins University 2019 – 2020
  - Assisted instruction in image analysis, mathematical modelling, and algorithmic problem solving.

## Mentoring

- Mentored undergraduate and postgraduate students at Johns Hopkins University and the University of Edinburgh.
- Mentees: Felipe Takaesu, Eliana Crentsil, Lucia Sablich, Shannon Flanary, and Chunhan Fang.

## Professional Service

- Reviewer, ACM Digital Threats: Research and Practice (DTRAP).
- Reviewer, AAAI Conference on Artificial Intelligence (AAAI-26).
- Reviewer, ACM Workshop on Artificial Intelligence and Security (AISec), 2026.
- Reviewer, Privacy Enhancing Technologies Symposium (PETS), 2027.

## Awards, Fellowships, and Grants

- IEEE SaTML Conference Attendance Grant 2024, 2025, 2026
- PhD Full Scholarship, University of Edinburgh 2023 – 2026
- Division Achievement Award, Fast Stable Diffusion on Intel CPU, Intel AIA 2023
- Division Recognition Award, Neural Coder Partnering Alibaba Cloud, Intel CESG SW AI 2023
- Division Achievement Award, Innovation of Neural Coder, Intel AIA 2022
- Departmental Research Fellowship, Johns Hopkins University 2017
- Outstanding Graduate of the Year, Fudan University 2017
- JASSO Full Scholarship for Exchange Students, Japanese Government 2014

## Technical Skills

- **Programming & AI tools:** Python, MATLAB, C++, Java, PyTorch, TensorFlow, NumPy/Pandas, Linux/Unix, Git, LaTeX, Codex app.
- **ML and computer vision:** deep learning, CNNs, U-Net, diffusion models, GANs, VAEs, image forensics, microscopy image analysis, model training and evaluation.
- **AI security and trust:** model attribution, provenance, fingerprinting, watermarking, adversarial evaluation, threat modelling, privacy and fairness measurement.
- **ML systems:** Intel Extension for PyTorch, Intel Neural Compressor, Neural Coder, INT8 quantization, model compression, inference optimization, benchmarking.
- **Research practice:** experimental design, benchmark construction, ablation studies, statistical thresholding, ROC/AUROC/FPR/TPR analysis, reproducibility.
- **Communication:** academic writing, technical reports, conference presentations, guest lectures, interdisciplinary collaboration, partner-facing communication.

## Languages and International Experience

- Chinese: native speaker.
- English: professional academic working proficiency, developed through full-time master's study in the United States and full-time PhD study in the United Kingdom.
- Japanese: JLPT N1; one-year full-time undergraduate exchange at Osaka University with JASSO scholarship support.

## References

Available upon request.